# A Practical Solution to the Multiple-Testing Crisis in Financial Research

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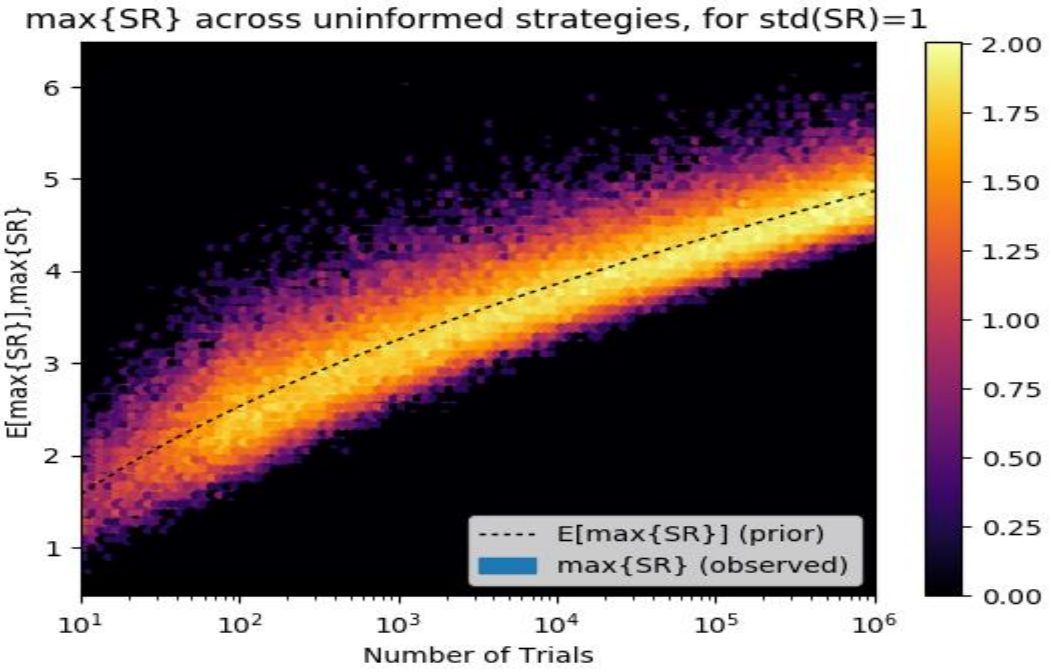
# Key Points

- Most discoveries in empirical finance are false, as a consequence of selection bias under multiple testing.
- This may explain why so many hedge funds fail to perform as advertised or as expected, particularly in the quantitative space.
- These false discoveries may have been prevented if academic journals and investors demanded that any reported investment performance incorporates the false positive probability, adjusted for selection bias under multiple testing.
- In this presentation, we demonstrate how this adjusted false positive probability can be computed and reported for public consumption.

The full paper can be downloaded at https://ssrn.com/abstract=3177057

# Section I
# The Problem

# The Most Important Plot In Finance




The y-axis displays the distribution of the maximum Sharpe ratios (max{SR}) for a given number of trials (x-axis). A lighter color indicates a higher probability of obtaining that result, and the dash-line indicates the expected value. For example, after only 1,000 independent backtests, the expected maximum Sharpe ratio (E[max{SR}]) is 3.26, even if the true Sharpe ratio of the strategy is zero!

The reason is *Backtest Overfitting*: When selection bias (picking the best result) takes place under multiple testing (running many alternative configurations), that backtest is likely to be a false discovery. **Most quantitative firms invest in false discoveries.**

# The "False Strategy" Theorem [2014]

- Given a sample of IID-Gaussian Sharpe ratios, $\left\{\widehat{SR}_k\right\}$, $k = 1, \ldots, K$, with $\widehat{SR}_k \sim \mathcal{N}\left[0, \mathrm{V}\left[\left\{\widehat{SR}_k\right\}\right]\right]$, then

$$\mathrm{E}\left[\max_k \left\{\widehat{SR}_k\right\}\right]\left(\mathrm{V}\left[\left\{\widehat{SR}_k\right\}\right]\right)^{-1/2} \approx$$
$$(1-\gamma)Z^{-1}\left[1-\frac{1}{K}\right]+\gamma Z^{-1}\left[1-\frac{1}{Ke}\right]$$

where $Z^{-1}[.]$ is the inverse of the standard Gaussian CDF, $e$ is Euler's number, and $\gamma$ is the Euler-Mascheroni constant.

- Corollary: Unless $\max_k\left\{\widehat{SR}_k\right\} \gg \mathrm{E}\left[\max_k\left\{\widehat{SR}_k\right\}\right]$, the discovered strategy is likely to be a *false positive*.

Source: López de Prado et al. (2014): "The effects of backtest overfitting on out-of-sample performance." *Notices of the American Mathematical Society, 61(5)*, pp. 458-471.

# Section II
# The Solution

# The Deflated Sharpe Ratio (1/2)

- The Deflated Sharpe Ratio computes the probability that the Sharpe Ratio (SR) is statistically significant, after controlling for the inflationary effect of multiple trials, data dredging, non-normal returns and shorter sample lengths.

$$\widehat{DSR} \equiv \widehat{PSR}\left(\widehat{SR}_0\right) = Z\left[\frac{\left(\widehat{SR} - \widehat{SR}_0\right)\sqrt{T-1}}{\sqrt{1 - \hat{\gamma}_3 \widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4}\widehat{SR}^2}}\right]$$

where

$$\widehat{SR}_0 = \sqrt{V\left[\left\{\widehat{SR}_k\right\}\right]}\left((1-\gamma)Z^{-1}\left[1 - \frac{1}{K}\right] + \gamma Z^{-1}\left[1 - \frac{1}{Ke}\right]\right)$$

- DSR packs more information than SR, and it is expressed in probabilistic terms.

# The Deflated Sharpe Ratio (2/2)

- The standard SR is computed as a function of two estimates:
  - Mean of returns
  - Standard deviation of returns.
- DSR deflates SR by taking into consideration five additional variables (it packs more information):
  - The non-Normality of the returns $(\hat{\gamma}_3, \hat{\gamma}_4)$
  - The length of the returns series $(T)$
  - The amount of data dredging $\left(\mathrm{V}\left[\left\{\widehat{SR}_k\right\}\right]\right)$
  - The number of independent trials involved in the discovered strategy $(K)$

DSR can be used to determine the probability that a discovered strategy is a **False Positive**. The key is to record all trials, and determine correctly the clusters of effectively independent trials.

# Section III
# A Numerical Example

# Selected Strategy

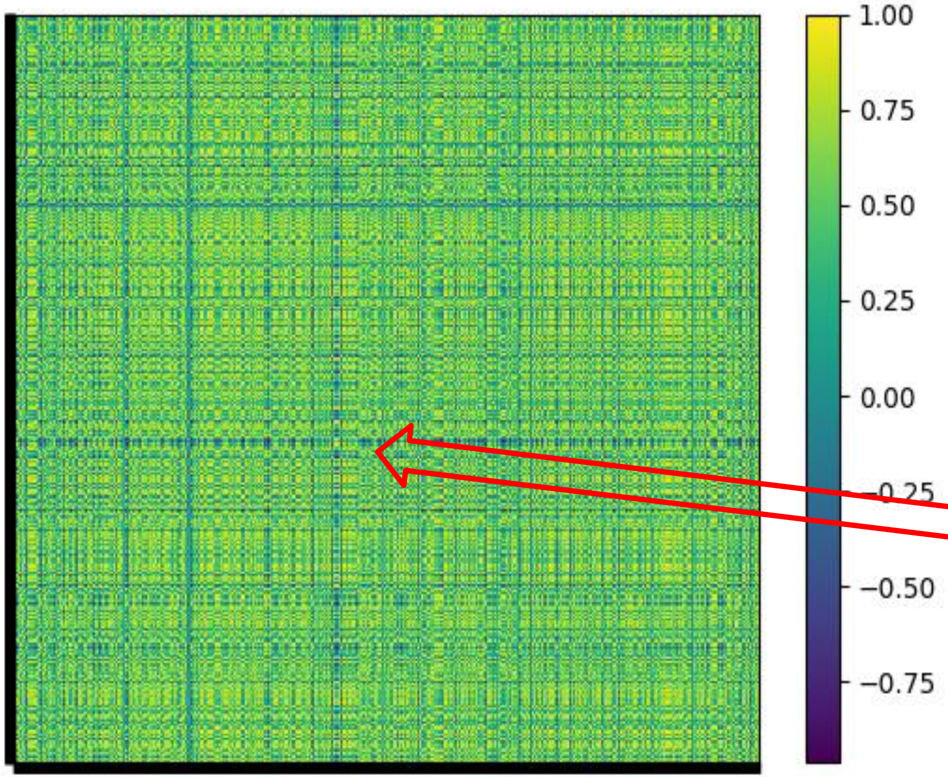
| Statistic | Strategy |
|---|---|
| Start date | 1/21/2010 |
| End date | 5/1/2018 |
| aRoR (Total) | 9.35% |
| Avg AUM (1E6) | 1506.43 |
| Avg Gini | 0.88 |
| Avg Duration | 0.08 |
| Avg Default Prob | 1.58% |
| An. Sharpe ratio | 2.00 |
| Turnover | 5.68 |
| Efficient Number | 186.26 |
| Correl to Ix | 0.48 |
| Drawdown (95%) | 2.89% |
| Time Underwater (95%) | 0.20 |
| Leverage | 3.59 |

Consider a long-short investment strategy on high grade corporate bonds, with the performance statistics reported on this table.

That annualized Sharpe ratio is 2.00, with an annualized return of 9.35% and an average duration of only 0.08. The 95-percentile of the drawdowns is only 2.89%, and the 95-percentile of the time underwater is only 0.2 years.

Most investors would agree that this appears to be a good investment strategy…

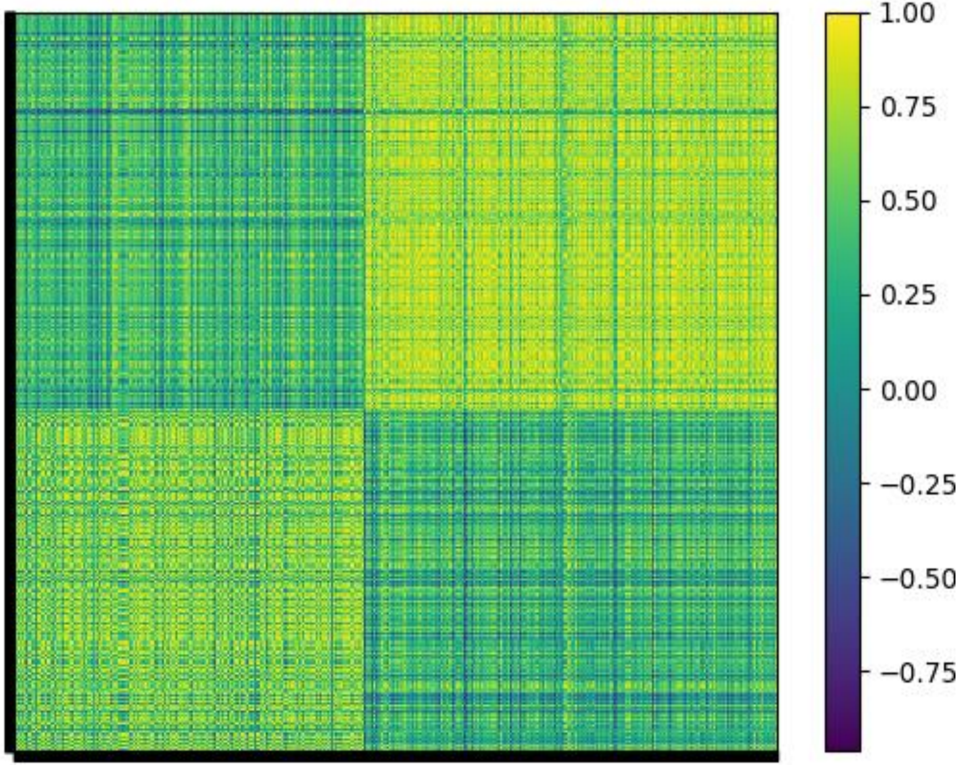
# Disclosure of All Trials




The problem is, this is the best strategy out of 6,385 trials conducted by the investment firm on this investment universe. The left plot represents the heat map of the correlations between trials' return series.

| Statistic | iBoxxIG | Strategy |
|---|---|---|
| Start date | 1/21/2010 | 1/21/2010 |
| End date | 5/1/2018 | 5/1/2018 |
| aRoR (Total) | 4.90% | 9.35% |
| Avg AUM (1E6) | 1000.00 | 1506.43 |
| Avg Gini | 0.29 | 0.88 |
| Avg Duration | 7.88 | 0.08 |
| Avg Default Prob | 1.36% | 1.58% |
| An. Sharpe ratio | 0.99 | 2.00 |
| Turnover | 0.64 | 5.68 |
| Efficient Number | 1034.87 | 186.26 |
| Correl to Ix | 1.00 | 0.48 |
| Drawdown (95%) | 3.17% | 2.89% |
| Time Underwater (95%) | 0.23 | 0.20 |
| Leverage | 1.00 | 3.59 |

# Clustering of Trials (1/10)



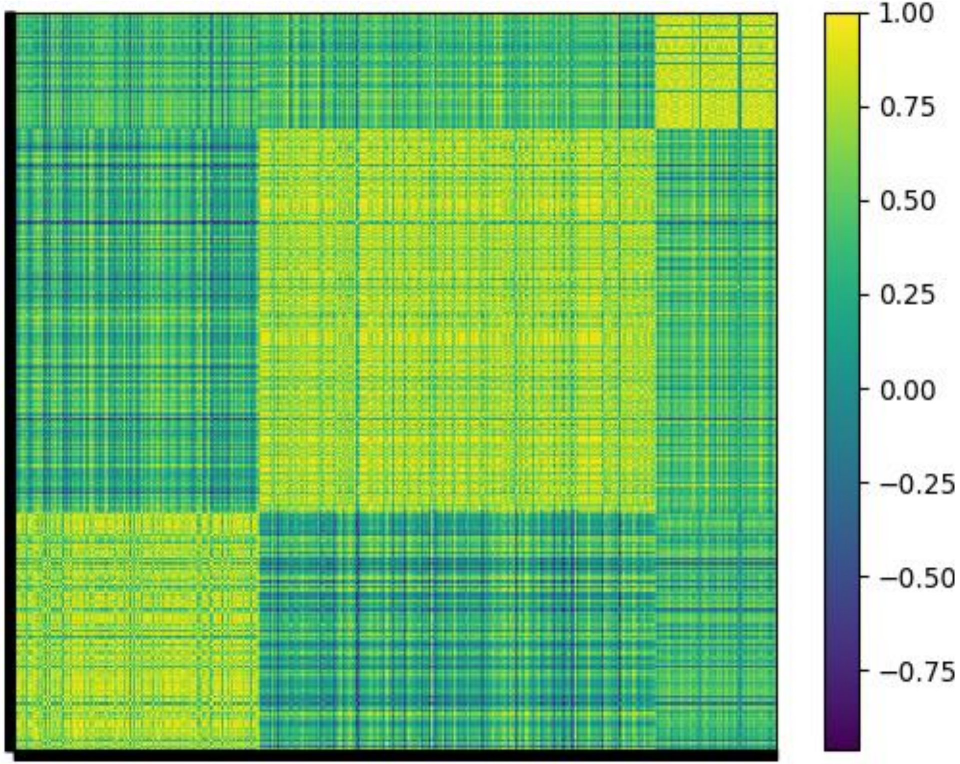

Fortunately, the 6,385 trials are not independent. The left plot represents the heat map of the correlations between trials' returns, grouped into 2 clusters.

The *quality* of this clustering, measured in terms of

$$q = \frac{\mathrm{E}[\{S_i\}]}{\sqrt{\mathrm{V}[\{S_i\}]}} = 2.3274$$

where $S_i$ is the silhouette score of bond *i*.

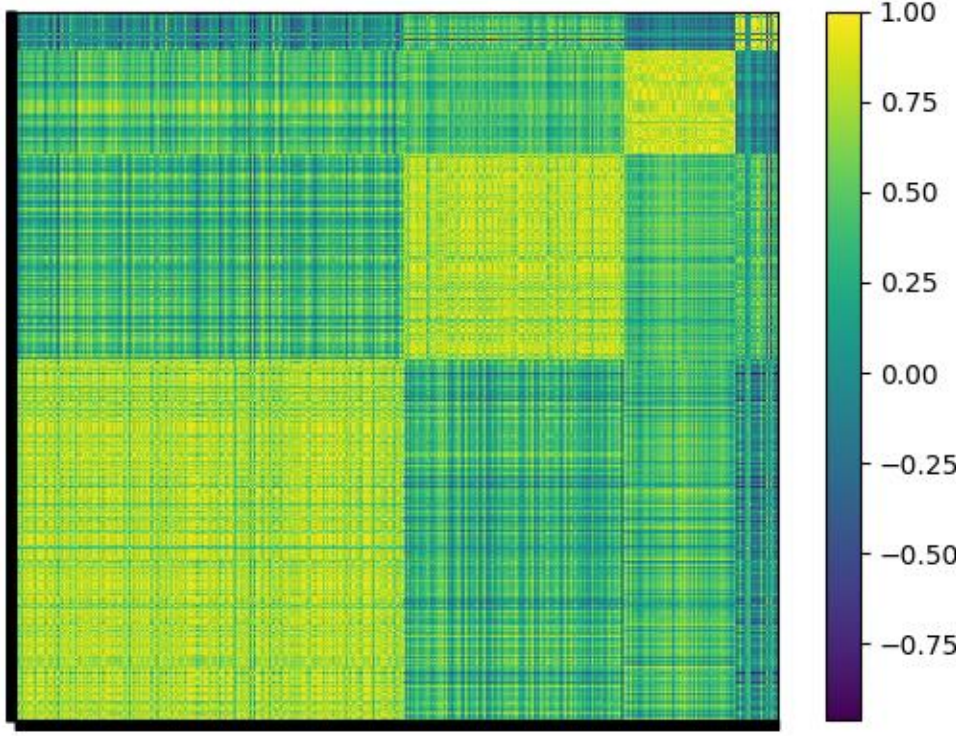
# Clustering of Trials (2/10)




When we split the trials into 3 clusters, the quality of the clustering increases from $q = 2.3274$ to $q = 2.7068$.

Interestingly, what happens is that one of the two previous clusters is further split into two, leading to a greater contrast between the block diagonal color and the off-diagonal colors.

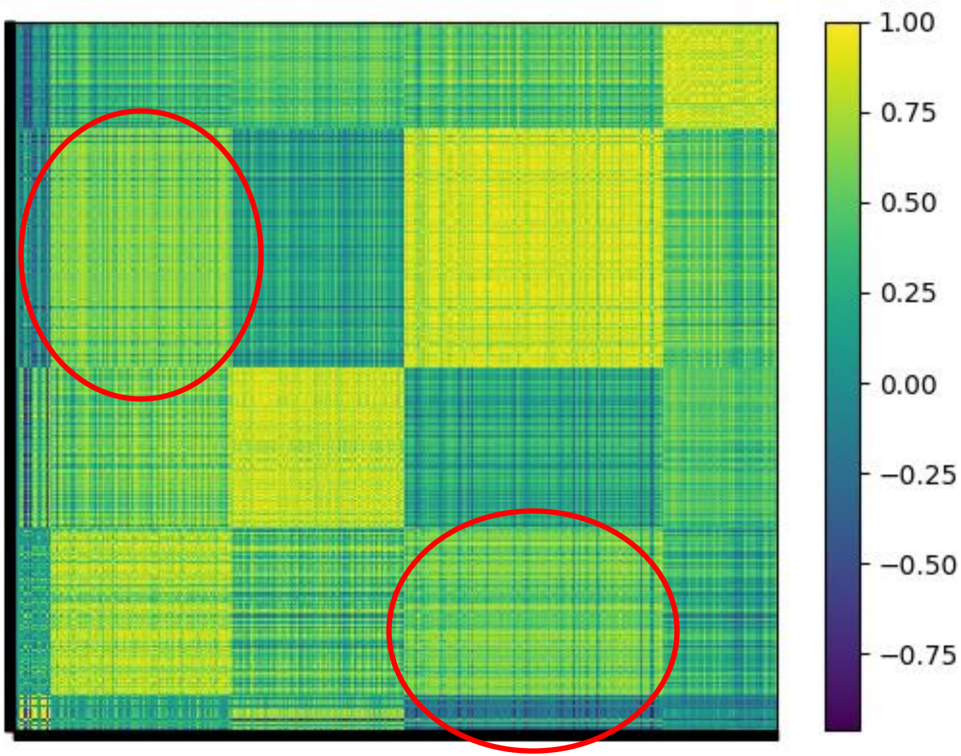
# Clustering of Trials (3/10)




When we split the trials into 4 clusters, the quality of the clustering increases from $q = 2.7068$ to $q = 2.7281$, a relatively minor improvement.

Out of the original two clusters, one has now been split into three.

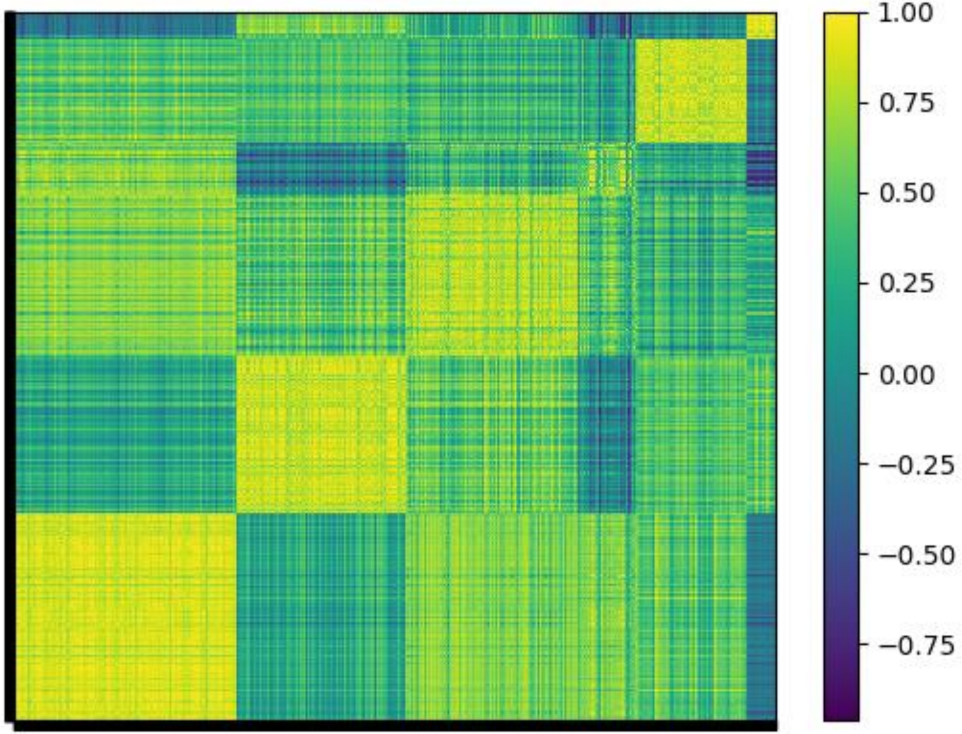
# Clustering of Trials (4/10)




When we split the trials into 5 clusters, the quality of the clustering decreases from $q = 2.7281$ to $q = 2.6517$.

This clustering quality is worse than what we observed for 4 and 3 clusters. This worsening can be visualized in the lighter off-diagonal blocks, and it is the consequence of splitting one of the two initial clusters into 2.

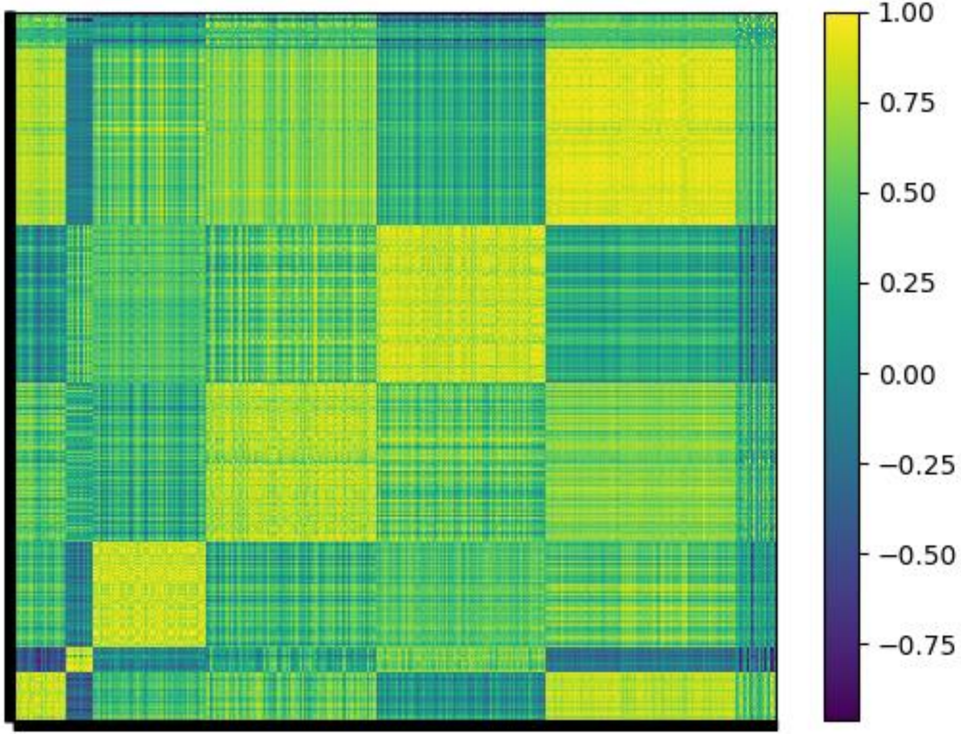
# Clustering of Trials (5/10)




When we split the trials into 6 clusters, the quality of the clustering <u>decreases</u> from $q = 2.6517$ to $q = 2.4919$.

Now we can observed multiple light-colored off-diagonal clusters.

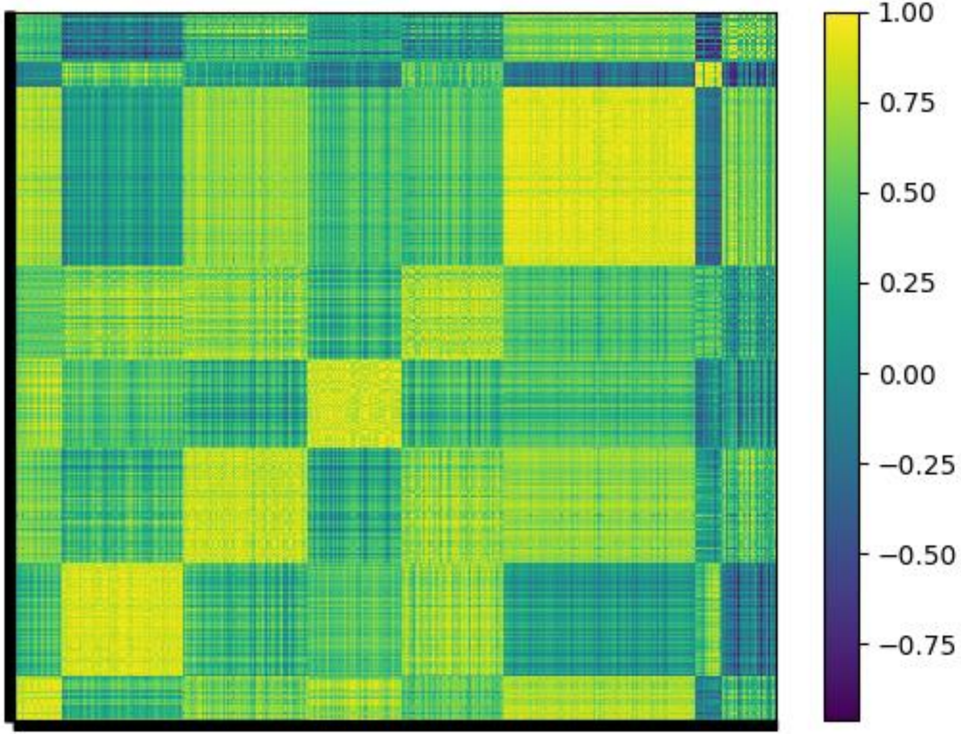
# Clustering of Trials (6/10)




When we split the trials into 7 clusters, the quality of the clustering <u>decreases</u> from $q = 2.4919$ to $q = 2.3650$.

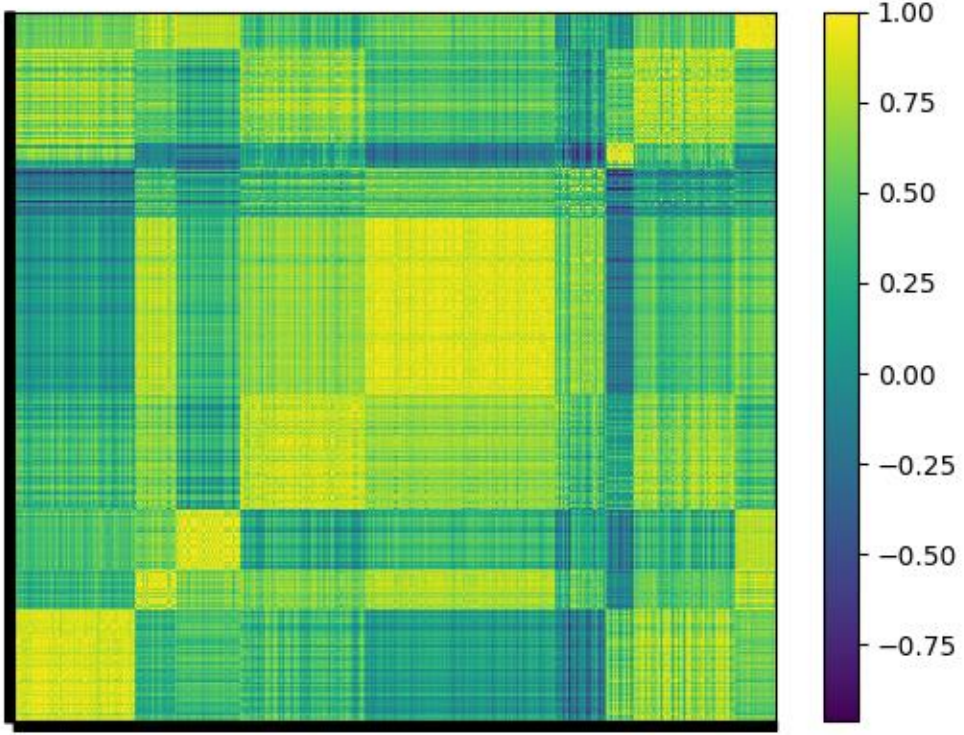
This clustering is only slightly better than the one with two clusters.

# Clustering of Trials (7/10)



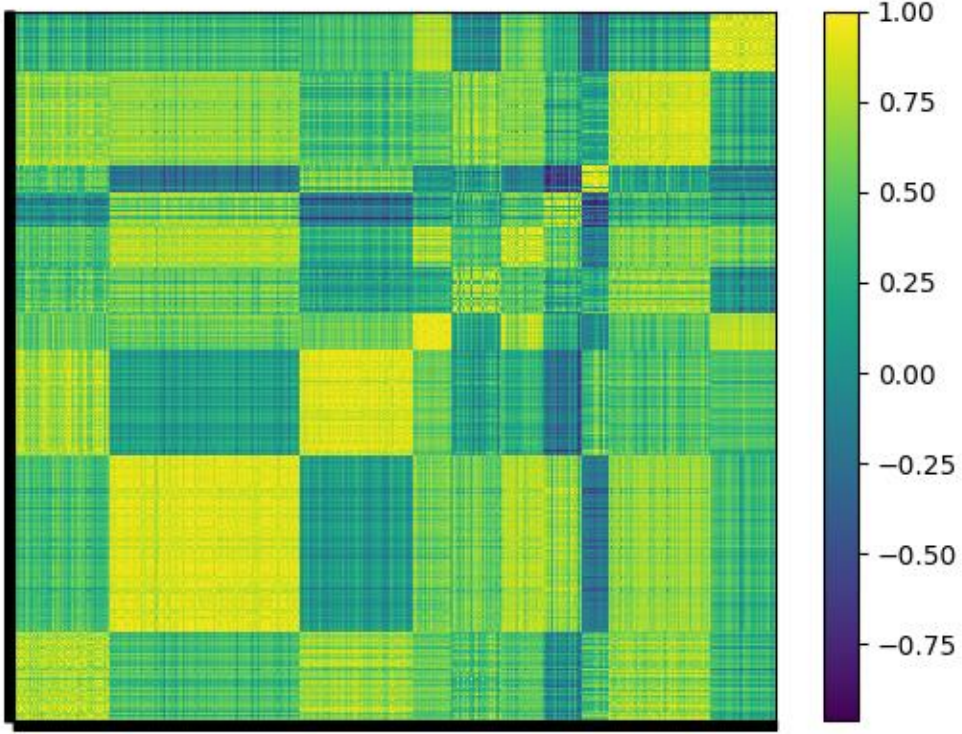

When we split the trials into 8 clusters, the quality of the clustering decreases from $q = 2.3650$ to $q = 2.2822$.

# Clustering of Trials (8/10)



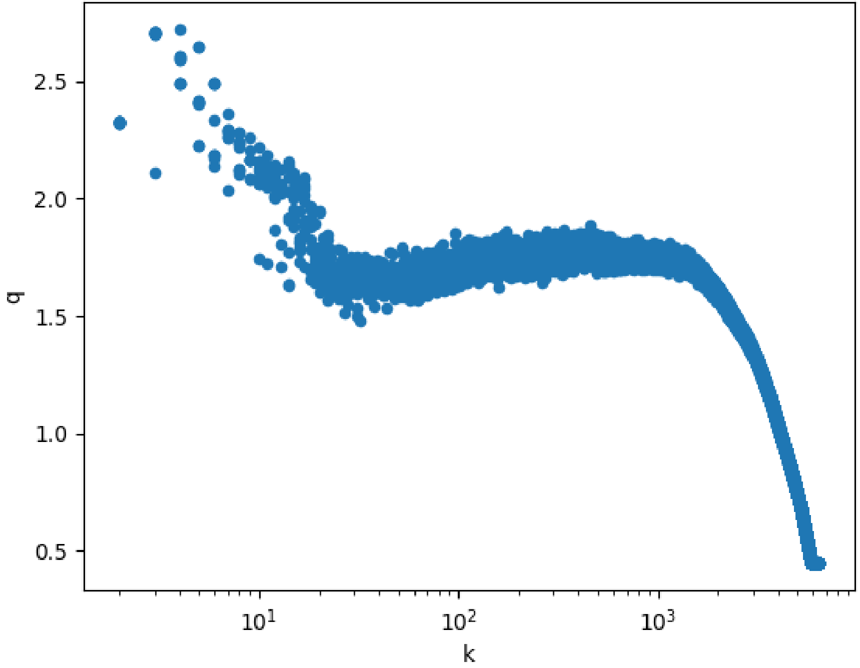

When we split the trials into 9 clusters, the quality of the clustering <u>decreases</u> from $q = 2.2822$ to $q = 2.2594$.

# Clustering of Trials (9/10)




When we split the trials into 10 clusters, the quality of the clustering decreases from $q = 2.2594$ to $q = 2.2211$.

# Clustering of Trials (10/10)




The scatter plot shows the clustering quality (y-axis) as we increase the number of clusters (x-axis, in polynomial scale) from 2 to 6,384.

As we observed earlier, the maximum quality is achieved for 4 clusters. Clustering quality decays abruptly after 10 clusters.

In conclusion, the firm has conducted 4 substantially uncorrelated trials on this investment universe.

# False Positive Probability

| Stats | Cluster 0 | Cluster 1 | Cluster 2 | Cluster 3 |
|---|---|---|---|---|
| **Strat Count** | 3265 | 1843 | 930 | 347 |
| **aSR** | 1.5733 | 1.4907 | 2.0275 | 1.0158 |
| **SR** | 0.0974 | 0.0923 | 0.1255 | 0.0629 |
| **Skew** | -0.3333 | -0.4520 | -0.4194 | 0.8058 |
| **Kurt** | 11.2773 | 6.0953 | 7.4035 | 14.2807 |
| **T** | 2172 | 2168 | 2174 | 2172 |
| **StartDt** | 2010-01-04 | 2010-01-04 | 2010-01-04 | 2010-01-04 |
| **EndDt** | 2018-05-01 | 2018-04-25 | 2018-05-03 | 2018-05-01 |
| **Freq** | 261.0474 | 261.0821 | 261.1159 | 261.0474 |
| **sqrt(V[SR_k])** | 0.0257 | 0.0256 | 0.0256 | 0.0257 |
| **E[max SR_k]** | 0.0270 | 0.0270 | 0.0270 | 0.0270 |
| **DSR** | 0.9993 | 0.9985 | **1.0000** | 0.9558 |

For each cluster, we can compute the returns associated with an inverse variance allocation to the trials involved. Then we can compute the performance statistics *per cluster*. This prevents that outliers may significantly bias results. The strategy selected in slide 10 belongs to Cluster 2, which contains 930 trials. The non-annualized Sharpe ratio is 0.1255.

For 4 clusters with a standard deviation across cluster SRs of 0.0256, the “False Strategy” theorem expects a maximum SR of 0.027. Taking into account the number of cluster returns, their skewness and kurtosis, the probability that the estimated SR is significantly greater than E[max{SR}] (a.k.a. DSR) is almost 1. **The probability of a false positive is virtually zero**.

# Robustness of Results (1/4)

| Stats | Cluster 0 | Cluster 1 |
|---|---|---|
| Strat Count | 2937 | 3448 |
| aSR | 1.7707 | 1.6023 |
| SR | 0.1096 | 0.0992 |
| Skew | -0.5780 | -0.3351 |
| Kurt | 6.5878 | 11.3212 |
| T | 2174 | 2172 |
| StartDt | 2010-01-04 | 2010-01-04 |
| EndDt | 2018-05-03 | 2018-05-01 |
| Freq | 261.1159 | 261.0474 |
| sqrt(V[SR_k]) | 0.0074 | 0.0074 |
| E[max SR_k] | 0.0038 | 0.0038 |
| DSR | **1.0000** | 1.0000 |

| Stats | Cluster 0 | Cluster 1 | Cluster 2 |
|---|---|---|---|
| Strat Count | 2063 | 3329 | 993 |
| aSR | 1.4411 | 1.5780 | 2.0638 |
| SR | 0.0892 | 0.0977 | 0.1277 |
| Skew | -0.4310 | -0.3357 | -0.4137 |
| Kurt | 5.8606 | 11.2267 | 7.3681 |
| T | 2170 | 2172 | 2174 |
| StartDt | 2010-01-04 | 2010-01-04 | 2010-01-04 |
| EndDt | 2018-04-27 | 2018-05-01 | 2018-05-03 |
| Freq | 261.1507 | 261.0474 | 261.1159 |
| sqrt(V[SR_k]) | 0.0202 | 0.0203 | 0.0202 |
| E[max SR_k] | 0.0173 | 0.0173 | 0.0173 |
| DSR | 0.9995 | 0.9999 | **1.0000** |

We can analyze the robustness of the previous result by repeating the calculations under alternative clusterings. For 2 clusters, the selected strategy belongs to Cluster 0, where the non-annualized SR is 0.1096. The expected maximum SR is 0.0038, and the DSR is again virtually 1.

For 3 clusters, the selected strategy belongs to Cluster 2, where the non-annualized SR is 0.1277. The expected maximum SR is 0.0173, and the DSR is again virtually 1.

# Robustness of Results (2/4)

| Stats | Cluster 0 | Cluster 1 | Cluster 2 | Cluster 3 | Cluster 4 |
|---|---|---|---|---|---|
| Strat Count | 317 | 1524 | 1434 | 2169 | 941 |
| aSR | 0.9690 | 1.4664 | 1.4065 | 1.5272 | 2.0319 |
| SR | 0.0600 | 0.0907 | 0.0870 | 0.0945 | 0.1257 |
| Skew | 2.2161 | -0.3286 | -0.4864 | -0.4086 | -0.4172 |
| Kurt | 41.2726 | 9.7988 | 5.4162 | 12.1809 | 7.4552 |
| T | 2172 | 2170 | 2168 | 2172 | 2174 |
| StartDt | 2010-01-04 | 2010-01-04 | 2010-01-04 | 2010-01-04 | 2010-01-04 |
| EndDt | 2018-05-01 | 2018-04-27 | 2018-04-25 | 2018-05-01 | 2018-05-03 |
| Freq | 261.0474 | 261.1507 | 261.0821 | 261.0474 | 261.1159 |
| sqrt(V[SR_k]) | 0.0234 | 0.0234 | 0.0234 | 0.0234 | 0.0234 |
| E[max SR_k] | 0.0279 | 0.0279 | 0.0279 | 0.0279 | 0.0279 |
| DSR | 0.9418 | 0.9979 | 0.9964 | 0.9987 | **1.0000** |

For 5 clusters, the selected strategy belongs to Cluster 4, where the non-annualized SR is 0.1257. The expected maximum SR is 0.0279, and the DSR is again virtually 1.

| Stats | Cluster 0 | Cluster 1 | Cluster 2 | Cluster 3 | Cluster 4 | Cluster 5 |
|---|---|---|---|---|---|---|
| Strat Count | 1873 | 1418 | 1447 | 476 | 935 | 236 |
| aSR | 1.5205 | 1.4034 | 1.4580 | 1.3853 | 2.0296 | 0.4322 |
| SR | 0.0941 | 0.0869 | 0.0902 | 0.0857 | 0.1256 | 0.0267 |
| Skew | -0.4254 | -0.4872 | -0.3458 | 0.5432 | -0.4188 | 0.1344 |
| Kurt | 13.0185 | 5.4077 | 9.9281 | 16.1401 | 7.4308 | 5.6976 |
| T | 2170 | 2168 | 2170 | 2172 | 2174 | 2170 |
| StartDt | 2010-01-04 | 2010-01-04 | 2010-01-04 | 2010-01-04 | 2010-01-04 | 2010-01-04 |
| EndDt | 2018-04-27 | 2018-04-25 | 2018-04-27 | 2018-05-01 | 2018-05-03 | 2018-04-27 |
| Freq | 261.1507 | 261.0821 | 261.1507 | 261.0474 | 261.1159 | 261.1507 |
| sqrt(V[SR_k]) | 0.0321 | 0.0321 | 0.0321 | 0.0321 | 0.0321 | 0.0321 |
| E[max SR_k] | 0.0417 | 0.0418 | 0.0417 | 0.0418 | 0.0417 | 0.0417 |
| DSR | 0.9909 | 0.9797 | 0.9862 | 0.9807 | **0.9999** | 0.2421 |

For 6 clusters, the selected strategy belongs to Cluster 4, where the non-annualized SR is 0.1256. The expected maximum SR is 0.0417, and the DSR is again virtually 1.

# Robustness of Results (3/4)

| Stats | Cluster 0 | Cluster 1 | Cluster 2 | Cluster 3 | Cluster 4 | Cluster 5 | Cluster 6 |
|---|---|---|---|---|---|---|---|
| Strat Count | 443 | 232 | 940 | 1436 | 1418 | 1591 | 325 |
| aSR | 1.4985 | 0.4229 | 2.0314 | 1.4566 | 1.4034 | 1.4816 | 1.2380 |
| SR | 0.0927 | 0.0262 | 0.1257 | 0.0901 | 0.0869 | 0.0917 | 0.0766 |
| Skew | -0.4098 | 0.1355 | -0.4174 | -0.3447 | -0.4872 | -0.4488 | 10.2898 |
| Kurt | 10.4565 | 5.6820 | 7.4499 | 9.9064 | 5.4077 | 13.8743 | 295.3934 |
| T | 2170 | 2170 | 2174 | 2169 | 2168 | 2170 | 2172 |
| StartDt | 2010-01-04 | 2010-01-04 | 2010-01-04 | 2010-01-04 | 2010-01-04 | 2010-01-04 | 2010-01-04 |
| EndDt | 2018-04-27 | 2018-04-27 | 2018-05-03 | 2018-04-26 | 2018-04-25 | 2018-04-27 | 2018-05-01 |
| Freq | 261.1507 | 261.1507 | 261.1159 | 261.1164 | 261.0821 | 261.1507 | 261.0474 |
| sqrt(V[SR_k]) | 0.0298 | 0.0298 | 0.0298 | 0.0298 | 0.0298 | 0.0298 | 0.0298 |
| E[max SR_k] | 0.0413 | 0.0413 | 0.0413 | 0.0413 | 0.0413 | 0.0413 | 0.0413 |
| DSR | 0.9901 | 0.2403 | **0.9999** | 0.9868 | 0.9807 | 0.9884 | 0.9799 |

| Stats | Cluster 0 | Cluster 1 | Cluster 2 | Cluster 3 | Cluster 4 | Cluster 5 | Cluster 6 | Cluster 7 |
|---|---|---|---|---|---|---|---|---|
| Strat Count | 411 | 1021 | 1037 | 794 | 846 | 1606 | 228 | 442 |
| aSR | 1.8643 | 1.3267 | 1.4133 | 1.9881 | 1.5228 | 1.4607 | 0.3817 | 1.3586 |
| SR | 0.1154 | 0.0821 | 0.0875 | 0.1230 | 0.0942 | 0.0904 | 0.0236 | 0.0841 |
| Skew | -0.2217 | -0.4884 | -0.3657 | -0.4156 | -0.3822 | -0.4481 | 0.1270 | 1.6051 |
| Kurt | 13.2850 | 5.1541 | 10.3922 | 6.7874 | 7.4346 | 12.7538 | 5.3075 | 34.8674 |
| T | 2170 | 2167 | 2169 | 2174 | 2168 | 2170 | 2170 | 2172 |
| StartDt | 2010-01-04 | 2010-01-05 | 2010-01-04 | 2010-01-04 | 2010-01-04 | 2010-01-04 | 2010-01-04 | 2010-01-04 |
| EndDt | 2018-04-27 | 2018-04-25 | 2018-04-26 | 2018-05-03 | 2018-04-25 | 2018-04-27 | 2018-04-27 | 2018-05-01 |
| Freq | 261.1507 | 261.0477 | 261.1164 | 261.1159 | 261.0821 | 261.1507 | 261.1507 | 261.0474 |
| sqrt(V[SR_k]) | 0.0298 | 0.0298 | 0.0298 | 0.0298 | 0.0298 | 0.0298 | 0.0298 | 0.0298 |
| E[max SR_k] | 0.0435 | 0.0435 | 0.0435 | 0.0435 | 0.0435 | 0.0435 | 0.0435 | 0.0435 |
| DSR | 0.9994 | 0.9606 | 0.9772 | **0.9998** | 0.9895 | 0.9829 | 0.1774 | 0.9754 |

For 7 clusters, the selected strategy belongs to Cluster 2, where the non-annualized SR is 0.1257. The expected maximum SR is 0.0413, and the DSR is again virtually 1.

For 8 clusters, the selected strategy belongs to Cluster 3, where the non-annualized SR is 0.1230. The expected maximum SR is 0.0435, and the DSR is again virtually 1.

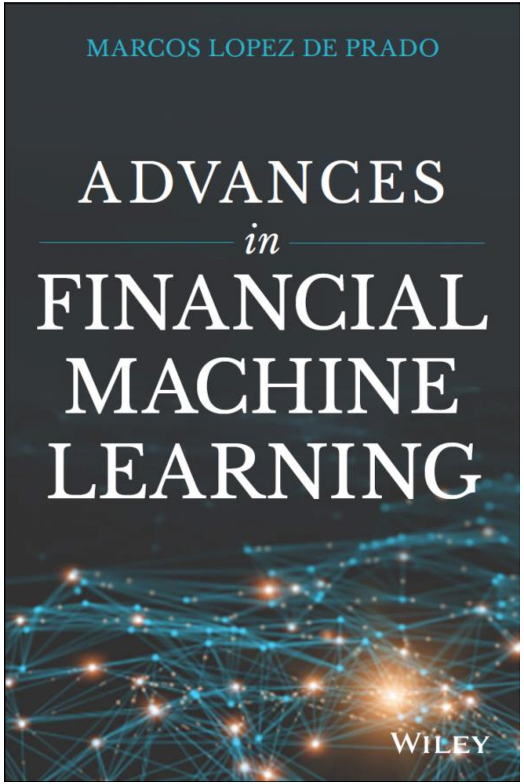
# Robustness of Results (4/4)

| Stats | Cluster 0 | Cluster 1 | Cluster 2 | Cluster 3 | Cluster 4 | Cluster 5 | Cluster 6 | Cluster 7 | Cluster 8 |
|---|---|---|---|---|---|---|---|---|---|
| **Strat Count** | 1021 | 352 | 536 | 1037 | 1593 | 440 | 228 | 846 | 332 |
| **aSR** | 1.3267 | 1.8185 | 1.8971 | 1.4133 | 1.4578 | 1.3482 | 0.3817 | 1.5228 | 1.9497 |
| **SR** | 0.0821 | 0.1125 | 0.1174 | 0.0875 | 0.0902 | 0.0834 | 0.0236 | 0.0942 | 0.1207 |
| **Skew** | -0.4884 | -0.2077 | -0.3769 | -0.3657 | -0.4467 | 2.2752 | 0.1270 | -0.3822 | -0.4008 |
| **Kurt** | 5.1541 | 13.3085 | 6.1852 | 10.3922 | 12.7629 | 49.3210 | 5.3075 | 7.4346 | 10.0715 |
| **T** | 2167 | 2170 | 2160 | 2169 | 2170 | 2172 | 2170 | 2168 | 2171 |
| **StartDt** | 2010-01-05 | 2010-01-04 | 2010-01-22 | 2010-01-04 | 2010-01-04 | 2010-01-04 | 2010-01-04 | 2010-01-04 | 2010-01-04 |
| **EndDt** | 2018-04-25 | 2018-04-27 | 2018-05-03 | 2018-04-26 | 2018-04-27 | 2018-05-01 | 2018-04-27 | 2018-04-25 | 2018-04-30 |
| **Freq** | 261.0477 | 261.1507 | 260.9792 | 261.1164 | 261.1507 | 261.0474 | 261.1507 | 261.0821 | 261.0131 |
| **sqrt(V[SR_k])** | 0.0290 | 0.0290 | 0.0290 | 0.0290 | 0.0290 | 0.0290 | 0.0290 | 0.0290 | 0.0290 |
| **E[max SR_k]** | 0.0441 | 0.0441 | 0.0441 | 0.0441 | 0.0441 | 0.0441 | 0.0441 | 0.0441 | 0.0441 |
| **DSR** | 0.9580 | 0.9990 | **0.9995** | 0.9755 | 0.9813 | 0.9736 | 0.1696 | 0.9886 | 0.9997 |

| Stats | Cluster 0 | Cluster 1 | Cluster 2 | Cluster 3 | Cluster 4 | Cluster 5 | Cluster 6 | Cluster 7 | Cluster 8 | Cluster 9 |
|---|---|---|---|---|---|---|---|---|---|---|
| **Strat Count** | 806 | 1596 | 948 | 332 | 409 | 353 | 327 | 227 | 851 | 536 |
| **aSR** | 1.5222 | 1.4586 | 1.3083 | 1.9497 | 1.3378 | 1.8174 | 1.2172 | 0.3787 | 1.4057 | 1.8971 |
| **SR** | 0.0942 | 0.0903 | 0.0810 | 0.1207 | 0.0828 | 0.1125 | 0.0753 | 0.0234 | 0.0870 | 0.1174 |
| **Skew** | -0.3953 | -0.4461 | -0.4847 | -0.4008 | -0.1356 | -0.2065 | 4.5167 | 0.1274 | -0.4064 | -0.3769 |
| **Kurt** | 6.9109 | 12.7512 | 5.1189 | 10.0715 | 7.4999 | 13.3321 | 108.1831 | 5.3035 | 10.9871 | 6.1852 |
| **T** | 2168 | 2170 | 2167 | 2171 | 2170 | 2170 | 2172 | 2170 | 2169 | 2160 |
| **StartDt** | 2010-01-04 | 2010-01-04 | 2010-01-05 | 2010-01-04 | 2010-01-04 | 2010-01-04 | 2010-01-04 | 2010-01-04 | 2010-01-04 | 2010-01-22 |
| **EndDt** | 2018-04-25 | 2018-04-27 | 2018-04-25 | 2018-04-30 | 2018-04-27 | 2018-04-27 | 2018-05-01 | 2018-04-27 | 2018-04-26 | 2018-05-03 |
| **Freq** | 261.0821 | 261.1507 | 261.0477 | 261.0131 | 261.1507 | 261.1507 | 261.0474 | 261.1507 | 261.1164 | 260.9792 |
| **sqrt(V[SR_k])** | 0.0278 | 0.0278 | 0.0278 | 0.0279 | 0.0278 | 0.0278 | 0.0278 | 0.0278 | 0.0278 | 0.0279 |
| **E[max SR_k]** | 0.0438 | 0.0438 | 0.0439 | 0.0439 | 0.0438 | 0.0438 | 0.0439 | 0.0438 | 0.0438 | 0.0439 |
| **DSR** | 0.9889 | 0.9819 | 0.9544 | 0.9997 | 0.9636 | 0.9990 | 0.9483 | 0.1706 | 0.9748 | **0.9995** |

Same conclusion for 9 and 10 clusters. Beyond 10 clusters, the quality of the clustering is so poor that we can consider them unrealistic scenarios

# For Additional Details




*The first wave of quantitative innovation in finance was led by Markowitz optimization. Machine Learning is the second wave and it will touch every aspect of finance. López de Prado's Advances in Financial Machine Learning is essential for readers who want to be ahead of the technology rather than being replaced by it.*
— Prof. **Campbell Harvey**, Duke University. Former President of the American Finance Association.

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— Prof. **Frank Fabozzi**, EDHEC Business School. Editor of The Journal of Portfolio Management.

# THANKS FOR YOUR ATTENTION!

# Bio

Dr. Marcos López de Prado is the chief executive officer of *True Positive Technologies*. He founded *Guggenheim Partners*' Quantitative Investment Strategies (QIS) business, where he developed high-capacity machine learning (ML) strategies that consistently delivered superior risk-adjusted returns. After managing up to $13 billion in assets, Marcos acquired QIS and spun-out that business from Guggenheim in 2018.

Since 2010, Marcos has been a research fellow at *Lawrence Berkeley National Laboratory* (U.S. Department of Energy, Office of Science). One of the top-10 most read authors in finance (SSRN's rankings), he has published dozens of scientific articles on ML and supercomputing in the leading academic journals, and he holds multiple international patent applications on algorithmic trading.

Marcos earned a PhD in Financial Economics (2003), a second PhD in Mathematical Finance (2011) from *Universidad Complutense de Madrid*, and is a recipient of Spain's National Award for Academic Excellence (1999). He completed his post-doctoral research at *Harvard University* and *Cornell University*, where he teaches a Financial ML course at the School of Engineering. Marcos has an Erdős #2 and an Einstein #4 according to the *American Mathematical Society*.

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